

Class Activity 1 – MDP

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COURSE CODE:MLA0305

COURSE NAME:REINFORCEMENT LEARNING

R-CODE:

```
# Markov Decision Process
```

```
grid_size <- 4
```

```
start <- c(1,1)
```

```
goal <- c(4,4)
```

```
state <- start
```

```
# Function to draw coloured grid
```

```
drawGrid <- function(state, step){
```

```
  grid.newpage()
```

```
  grid.text(
```

```
    paste("Markov Decision Process - Step", step),
```

```
    y=0.97,
```

```
    gp=gpar(fontsize=18,fontface="bold")
```

```
  )
```

```
  for(i in 1:grid_size){
```

```
    for(j in 1:grid_size){
```

```
      fill <- "white"
```

```

if(i==start[1] && j==start[2])

    fill <- "skyblue"

if(i==goal[1] && j==goal[2])

    fill <- "lightgreen"

    if(i==state[1] && j==state[2])

        fill <- "red"

x <- (j-0.5)/grid_size

y <- 1-(i-0.5)/grid_size

grid.rect(

    x=x,

    y=y,

    width=1/grid_size,

    height=1/grid_size,

    gp=gpar(fill=fill,col="black",lwd=2)

)

if(i==start[1] && j==start[2])

    grid.text("S",x,y,gp=gpar(fontsize=18,fontface="bold"))

if(i==goal[1] && j==goal[2])

    grid.text("G",x,y,gp=gpar(fontsize=18,fontface="bold"))

if(i==state[1] && j==state[2])

    grid.text("A",x,y,

        gp=gpar(col="white",

            fontsize=18,

            fontface="bold"))

```

```
}  
}
```

```
# Legend
```

```
grid.text("Blue = Start  Red = Agent  Green = Goal",
```

```
        y=0.03,
```

```
        gp=gpar(fontsize=12,fontface="bold"))
```

```
}
```

```
# Simulation
```

```
step <- 0
```

```
total_reward <- 0
```

```
cat("=====\n")
```

```
cat(" MARKOV DECISION PROCESS SIMULATION\n")
```

```
cat("=====\n\n")
```

```
drawGrid(state, step)
```

```
cat("Step :", step, "\n")
```

```
cat("Current State :", state, "\n")
```

```
cat("Action : Start\n")
```

```
cat("Reward : 0\n\n")
```

```
Sys.sleep(1)
```

```
while(!all(state==goal)){
```

```
    current <- state
```

```
    if(state[1] < 4){
```

```
        state <- c(state[1]+1,state[2])
```

```

    action <- "Down"

  }else{

    state <- c(state[1],state[2]+1)

    action <- "Right"

  }

  step <- step + 1

  if(all(state==goal))

    reward <- 100

  else

    reward <- -1

  total_reward <- total_reward + reward

  # Visual Output

  drawGrid(state, step)

  # Console Summary

  cat("-----\n")

  cat("Step      :", step,"\n")

  cat("Current State :", current,"\n")

  cat("Action      :", action,"\n")

  cat("Next State   :", state,"\n")

  cat("Reward       :", reward,"\n\n")

  Sys.sleep(1)

}

cat("=====\n")

cat(" FINAL SUMMARY\n")

```

```
cat("=====\n")
```

```
cat("Start State :", start,"\n")
```

```
cat("Goal State :", goal,"\n")
```

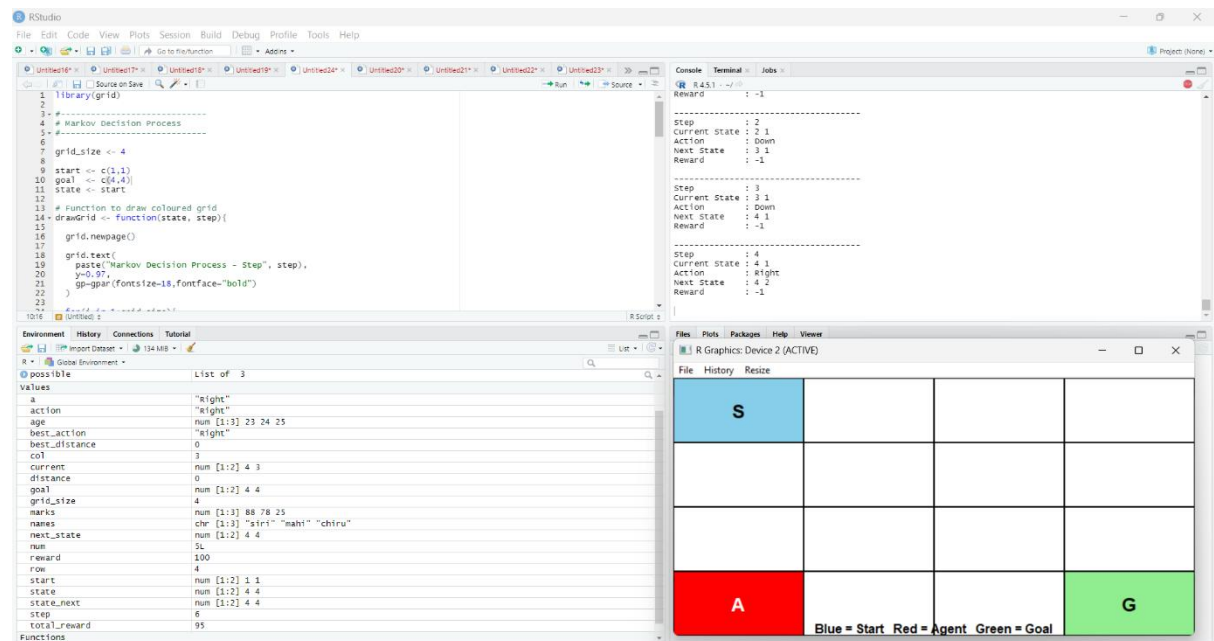
```
cat("Final State :", state,"\n")
```

```
cat("Total Steps :", step,"\n")
```

```
cat("Total Reward :", total_reward,"\n")
```

```
cat("Goal Reached : YES\n")
```

Full screenshot of Rstudio:



Visual

result:

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
1.00 [Source] [Go to Definition] [Add]
Console Terminal Jobs
R4.1.1 - r4.1.1
reward : -1
-----
Step : 2
Current State : 2 1
Action : down
Next State : 3 1
Reward : -1
-----
Step : 3
Current State : 3 1
Action : down
Next State : 4 1
Reward : -1
-----
Step : 4
Current State : 4 1
Action : right
Next State : 4 2
Reward : -1
-----
Step : 5
Current State : 4 2
Action : right
Next State : 4 3
Reward : -1
-----
Step : 6
Current State : 4 3
Action : right
Next State : 4 4
Reward : 100
>
> cat("=====\n")
> cat(" FINAL SUMMARY\n")
> cat("=====\n")
> cat("Start State : ", start,"\n")
> cat("Goal state : ", goal,"\n")
> cat("Final state : ", state,"\n")
> cat("Total steps : ", step,"\n")
> cat("Total Reward : ", total_reward,"\n")
> cat("Goal Reached : YES\n")
Files Plots Packages Help Viewer
```

Summary result:

R Graphics: Device 2 (ACTIVE)			
File History Resize			
S			
	A		G
Blue = Start. Red = Agent Green = Goal			